GEOMETRY · RIGHT TRIANGLES AND TRIGONOMETRY

# Trig Ratio Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Trig Ratio Explorer** on Mr. K's Classroom: Geometry → Right Triangles and Trigonometry → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

## 1 Predict

before you touch the tool

1. If you double a right triangle's size without changing its angles, does sin θ change?

## 2 Explore

use the tool and record what you see

**Ratios at different angles**

| θ | sin θ | cos θ | tan θ |
|---|---|---|---|
| 30° | | | |
| 45° | | | |
| 60° | | | |
| 35° | | | |

**Same angle (35°), different sizes**

| Hypotenuse | Opposite | Adjacent | sin 35° |
|---|---|---|---|
| 4 | | | |
| 8 | | | |
| 12 | | | |

**Special right triangles**

Use the second tab.

| Triangle | Short leg | Other leg | Hypotenuse |
|---|---|---|---|
| 45°-45°-90° | 3 | | |
| 30°-60°-90° | 4 | | |
| 30°-60°-90° | 5 | | |

## 3 Write the rule

fill in each blank from what you observed

1. $\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$, $\cos\theta =$ _____ and $\tan\theta =$ _____.

2. Changing the size of the triangle _____ the trig ratios, because the triangles are similar.

3. $\sin\theta = \cos($_____$)$.

4. In a 45°-45°-90° triangle, hypotenuse = leg × _____. In a 30°-60°-90° triangle, the long leg = short leg × _____.

## 4 Check no tool: show your work

1. A right triangle has legs 6 and 8. Find the hypotenuse and the sine of the angle across from the 6.

2. A tree casts a 20 ft shadow when the sun is 40° above the ground. How tall is the tree?

3. A 30°-60°-90° triangle has hypotenuse 14. Find both legs.

4. Solve $\sin 28° = \frac{x}{15}$.

## 5 Reflect

Why does every 35° right triangle have the same sine?